

ChristosHadjinikolis

Engineering Manager | Production ML & Streaming Systems

Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21

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mediums



organisation

org scale-up
hiring loops
mentoring/retention
cross-functional delivery

systems

client-facing ML
high-volume streaming
observability
reliability

engineering

operating models
streaming platforms
DORA/Jira signals
local E2E tests

technical

ML systems
Java/Kotlin
Python/SQL
Kafka/Flink
Airflow/AWS/GCP

public

JTC'21 standards
UCL research
PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

Engineering Manager for ML and real-time data systems with **16 years** across AI research, consulting, backend engineering, and live data platforms. Lead teams that turn noisy data, experimental models, and ambiguous product semantics into reliable, client-facing systems. At Vortexa, own a major **live streaming backend estate**, lead a **10-person cross-functional team**, and drove Data Production Team growth from **4 to 30+ people** by shaping hiring, system-design culture, platform choices, and operating discipline. My UCL and JTC'21 standards work informs how I treat auditability, explainability, versioning, and safe adoption in practice. I build engineering organisations, operating models, and platforms that keep ML systems dependable when product quality and client trust are on the line.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own the **engineering strategy and delivery** for Vortexa's **live streaming intelligence estate**, a major client-facing backend area powering product and commercial workflows, filtering roughly **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Lead **live-processing strategy** with a **10-person cross-functional team** spanning **6 MLE/DE/DS, 2 Product, and 2 SMEs**. Because this estate feeds live product behaviour, reliability, explainability, monitoring, and response readiness are delivery requirements.

- **People & Org Scaling:** Directly manage the MLE/DE/DS core and lead a **10-person cross-functional delivery team**; drove Data Production Team growth from **4 to 30+** through hiring loops and system design evaluations; hiring manager for **6 roles**, interviewing **31+ candidates** in that capacity and **60+ overall**; retained the full current team, with every member recently promoted.
- **Platform Scaling:** Enabled independent scaling of critical live pipelines by leading the two-year **Kafka Streams-to-Flink migration**; used Flink's dataflow model, independent state/checkpointing, richer operational UI, and framework-driven design to reduce maintenance effort.
- **Trustworthy ML Estate:** Established Vortexa's ML development lifecycle from research to production: prototyping discipline, **MLflow**, model/data versioning, service-based model deployment, replay/evaluation interfaces, and live accuracy-monitoring dashboards.
- **Reliability & Ops Culture:** Protect customer-facing reliability with monitoring, runbooks, rollback/fallback plans, and Jira alert workflows; moved from firefighter patterns to shared on-call ownership while keeping **MTTR under 30 min**.
- **Product Semantics & Interfaces:** Resolve recurring prediction-quality disputes through shared semantics, turning stakeholder friction into measurable evaluation targets and RFC/workstream-ready product interfaces.
- **Engineering Operating Model:** Scale delivery through Flink/Airflow archetypes, ADRs, dev containers, Backstage adoption, local E2E/integration tests, DORA/Jira signals, and a weekly architecture forum; latest view: **14.65 deploys/week** with **0.52% change failure**. Cut a **3h DAG feedback loop to under 5 min**.

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and played a formative role in growing it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; led client meetings, scoped goals, placed consultants, interviewed/mentored hires, and delivered ML/data work across UBS, Vodafone, CNHi and other enterprise clients. Ran the London Flink Meetup for five years.

- **Client Delivery Leadership:** Drove agendas with client stakeholders and product teams, agreed goals, facilitated technical delivery, and supported hiring/consultant placement across banking, telecoms, manufacturing, publishing, and gaming accounts.
- **Vodafone Group:** Led technical delivery across multiple workstreams; worked on Infinity, a GCP/**Kubeflow** DS platform, shaping ML lifecycle support; built Red Agent, a mobile-network feature-engineering framework.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance; managed/guided a DS team using **PySpark** over live vehicle sensory data with product-facing backlog discipline.
- **UBS:** Grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability work, and real-time pipelines with **Kafka**, **Elasticsearch**, and Python.

Roles in UK Institutions 2010–2016

Coding Teacher @DGC | Teaching Assistant @KCL/UCL

Built the communication and presentation discipline I now rely on as a leader by teaching Java/Python and core computer science at David Game College from 09/2012 to 04/2016 while completing doctoral research; Teaching Assistant at KCL and UCL.

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning, graph analytics , and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics. Received Best Poster Award at IJCAI 2013 (413 submissions). Awarded EPSRC PhD Scholarship , Outstanding TA Award (student-nominated), and earned the Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering with a strong focus on mathematics (10 courses), covering theoretical aspects, networking, and telecommunications. Majored in Artificial Intelligence .	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

committees/affiliations

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

selected public tooling

PyPI	dynamicio Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so ML pipeline logic can be tested with faster feedback.	ML/data I/O contracts
PyPI	skeleton-replay Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted understanding.	Runtime architecture evidence
JetBrains	Skeleton Replay plugin Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.	IDE workflow

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa , Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio , a published PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science explained by an ML expert, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

certificates & awards

07/2020	AWS Certified Machine Learning - Specialty Validation Number: D0R0WC7CN1VQ13SV	AWS
01/2019	Process Mining License number: AFU2VTP6273V	Coursera Course Certificates
10/2017	Google Certified Professional - Data Engineer Credential ID: 000000400	Google Cloud Certified
07/2017	Advanced Elasticsearch: Data modelling Enrolment Id: 33442	Elastic
06/2016	Graph Analytics for Big Data License number: NZADYAQ7N8FW	Coursera Course Certificates
05/2016	Apache Spark Developer Certification License number: 1.x-0548	O'Reilly Media
04/2016	Neo4j Certified Professional License number: here	Neo Technology, Creators of Neo4j